

CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title:

AI Assisted Emergency Department Triage Support System for Resource Constrained Healthcare Settings

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Problem Statement

Emergency department triage frequently depends on subjective clinical assessment, resulting in inconsistent patient prioritization and avoidable under triage of high acuity cases during periods of overcrowding. Recent literature demonstrates that AI assisted triage systems can improve prediction accuracy and decision consistency; however, most studies rely on retrospective, single site datasets with limited explainability and no reported prospective validation in Caribbean or small island healthcare contexts [1–5]. This project proposes evaluating an explainable machine learning decision support approach to improve triage consistency while maintaining clinician oversight.

Previous Work

[1] Da’Costa A, Teke J, Origbo JE, Osonuga A, Egbon E, Olawade DB. AI-driven triage in emergency departments: A review of benefits, challenges, and future directions. *Int J Med Inform.* 2025;197:105838.

This review examined how artificial intelligence is being applied to emergency department triage to reduce overcrowding and improve patient prioritization. AI was primarily used through machine learning and natural language processing approaches that processed patient data to support triage decisions and optimize resource allocation. The review reported improvements in wait times, reductions in time-to-treatment, and more efficient bed and staffing allocation. However, the authors identified challenges including algorithmic bias, clinician distrust, and concerns surrounding black-box decision making.

[2] Tyler S, Olis M, Aust N, Patel L, Simon L, Triantafyllidis C, Patel V, Lee DW, Ginsberg B, Ahmad H, Jacobs RJ. Use of artificial intelligence in triage in hospital emergency departments: A scoping review. *Cureus.* 2024;16(5):e59906.

This scoping review explored the use of artificial intelligence across emergency department triage systems to improve patient classification and outcome prediction. AI was applied using machine learning algorithms trained on patient demographics, symptoms, and clinical indicators to support prioritization decisions. The review found that AI systems frequently outperformed traditional triage approaches, with some models demonstrating substantially

improved predictive performance. The authors highlighted limitations related to retrospective datasets, inconsistent preprocessing methods and limited external validation.

[3] Araouchi Z, Adda M. TriageIntelli: AI-assisted multimodal triage system for health centers. *Procedia Comput Sci.* 2024;251:430–437.

This study developed TriageIntelli, an AI-assisted triage system designed to support emergency assessment and reduce delays caused by increasing patient volumes. Artificial intelligence was implemented through six supervised machine learning models trained on emergency department data and combined into a stacked ensemble architecture. The final stacked model achieved the highest performance, reaching 80.05% classification accuracy and an AUROC of 0.93. Despite strong performance, the study was limited by its reliance on historical data collected from only two Korean emergency departments.

[4] Chang et al. Using Machine Learning and Natural Language Processing in Triage for Prediction of Clinical Disposition in the Emergency Department. *BMC Emergency Medicine.* 2024.

This study investigated whether machine learning and natural language processing could improve emergency department triage by predicting patient clinical disposition. AI was used to analyze structured triage data together with free text clinical notes to identify patterns associated with admission outcomes. The results demonstrated improved predictive capability compared with conventional approaches and highlighted the value of combining structured and unstructured clinical information. Limitations included challenges associated with model generalization and dependence on local documentation practices.

[5] Masanneck et al. Triage Performance Across Large Language Models, ChatGPT, and Untrained Doctors in Emergency Medicine: Comparative Study. *J Med Internet Res.* 2024.

This comparative study evaluated the performance of large language models in emergency medicine triage tasks relative to inexperienced human decision makers. AI was implemented using large language models to generate triage recommendations across simulated emergency scenarios. The findings demonstrated promising performance and highlighted the potential of language models to support triage workflows and decision making. However, concerns remained regarding reliability, transparency and the safe deployment of generative AI systems in clinical environments.

Proposed Solution

This project proposes the development of an explainable AI assisted emergency triage decision support prototype that predicts patient acuity levels using structured clinical indicators and symptom information. Unlike existing approaches that prioritize prediction accuracy alone, the system will provide interpretable recommendations to support clinician judgement. The pilot will focus on evaluating whether explainable machine learning can improve consistency in triage classification while remaining feasible for resource constrained and Caribbean healthcare environments.

References

1. Da'Costa A, Teke J, Origbo JE, Osonuga A, Egbon E, Olawade DB. AI-driven triage in emergency departments: A review of benefits, challenges, and future directions. *Int J Med Inform.* 2025;197:105838.
2. Tyler S, Olis M, Aust N, Patel L, Simon L, Triantafyllidis C, Patel V, Lee DW, Ginsberg B, Ahmad H, Jacobs RJ. Use of artificial intelligence in triage in hospital emergency departments: A scoping review. *Cureus.* 2024;16(5):e59906.
3. Araouchi Z, Adda M. TriageIntelli: AI-assisted multimodal triage system for health centers. *Procedia Comput Sci.* 2024;251:430–437.
4. Chang et al. Using Machine Learning and Natural Language Processing in Triage for Prediction of Clinical Disposition in the Emergency Department. *BMC Emerg Med.* 2024.
5. Masanneck et al. Triage Performance Across Large Language Models, ChatGPT, and Untrained Doctors in Emergency Medicine: Comparative Study. *J Med Internet Res.* 2024.